Name: ______________________________ (please print)

Signature: ___________________________

ECE 2202 – Exam 1

September 13, 2025

Keep this exam closed until you are told to begin.

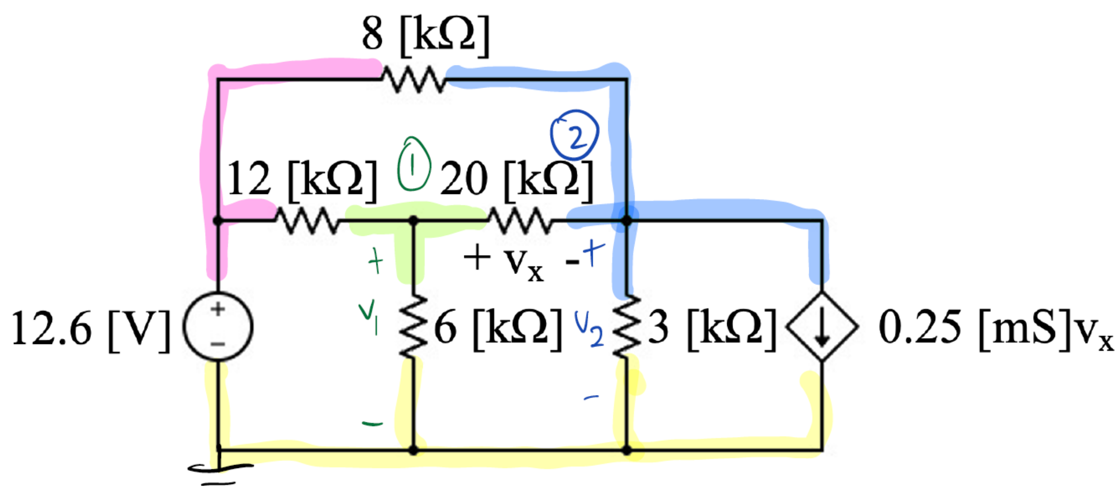
1. This exam is closed book, closed notes. A formula sheet is provided in the back.
2. Show all work on these pages. Show all work necessary to complete the problem. A solution without the appropriate work shown will receive no credit. A solution that is not given in a reasonable order will lose credit. Clearly indicate your answer (for example by enclosing it in a box).
3. Show all units in solutions, intermediate results, and figures. Units in the exam will be included between square brackets.
4. If the grader has difficulty following your work because it is messy or disorganized, you will lose credit.
5. Do not use red ink. Do not use red pencil.
6. You will have 100 minutes to work on this exam.

GRADE

1. ________________/25
2. ________________/25
3. ________________/25
4. ________________/25

Total = 100

1. {25 Points}Using nodal analysis (i.e., node voltage method),
   a. Label nodes needed to solve this problem as V_1, V_2, and GND, and write the two raw node voltage equations in the space provided.
   b. Simplify the two equations and write them in the space provided.
   c. Determine the values of V_1 and V_2.
   d. How much power is dissipated in the 3 [kΩ] resistor?

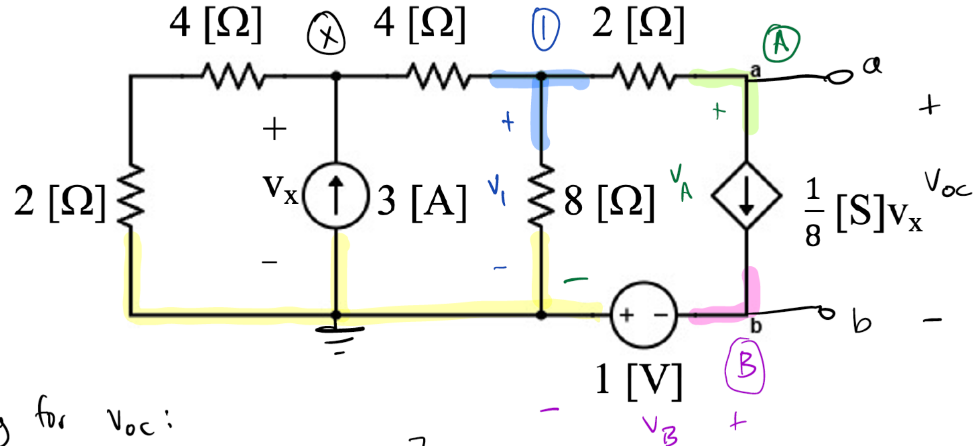


Raw equations $v_x = V_1 - V_2$	① $\left[\frac{V_1 - 12.6[V]}{12[k\Omega]} + \frac{V_1}{6[k\Omega]} + \frac{V_1 - V_2}{20[k\Omega]} = 0\right] \cdot 60$ ② $\left[\frac{V_2 - V_1}{20[k\Omega]} + \frac{V_2 - 12.6[V]}{8[k\Omega]} + \frac{V_2}{3[k\Omega]} + 0.25[mS]v_x = 0\right] \cdot 120$
Simplified equations	① $18V_1 - 3V_2 = 63 \Rightarrow 6V_1 - V_2 = 21$ ② $-6V_1 + 31V_2 + 30V_1 - 189 = 0 \Rightarrow 24V_1 + 31V_2 = 189$
V_1	$\boxed{V_1 = 4[V]}$
V_2	$\boxed{V_2 = 3[V]}$
P for 3 [k□] resistor	$P_{ABS,3[k\Omega]} = \frac{V_2^2}{3[k\Omega]} = \frac{3[V]^2}{3[k\Omega]} = 3[mW]$ $\boxed{P_{ABS,3[k\Omega]} = 3[mW]}$

Room for extra work

Units (2 pts), labeling (1 pt), organization (1 pt)
Raw equations (equations 12 pts)
Simplified equations (4 pts)
V1 correct answer with units (1 pt)
V2 correct answer with units (1 pt)
P equation correct (2 pts) and correct answer with units (1 pt)

{25 Points} Refer to the figure below.
a. Find the Thévenin equivalent circuit at terminals (a,b) in the circuit below.
b. If a load resistor is placed between terminals (a,b), what resistance value would ensure maximum power transfer?



a) solving for V_{oc}:

$$\text{(X)}\quad \frac{v_x}{4[\Omega]+2[\Omega]} + \frac{v_x - v_1}{4[\Omega]} - 3[A] = 0$$

$$\text{(1)}\quad \frac{v_1 - v_x}{4[\Omega]} + \frac{v_1}{8[\Omega]} + \frac{1}{8}[S]v_x = 0$$

$$\text{(A)}\quad \frac{v_A - v_1}{2[\Omega]} + \frac{1}{8}[S]v_x = 0$$

$\rightarrow$ $v_A = 3/4\ [V]$, $v_x = 9\ [V]$, $v_1 = 3\ [V]$

$v_B = -1\ [V]$

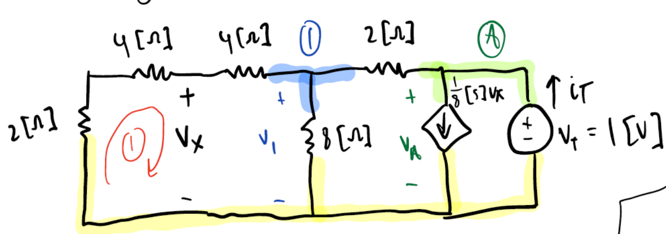
$V_{oc} = V_a - V_b$

$V_{oc} = \frac{3}{4}[V] - (-1[V])$

$\boxed{V_{oc} = 7/4\ [V]}$

Solving for Req:

$v_A = 1\ [V]$



$$\text{(1)}\quad \frac{v_1}{4[\Omega]+4[\Omega]+2[\Omega]} + \frac{v_1}{8[\Omega]} + \frac{v_1 - v_A}{2[\Omega]} = 0$$

$$\text{(A)}\quad \frac{v_A - v_1}{2[\Omega]} + \frac{1}{8}[S]v_x - i_T = 0$$

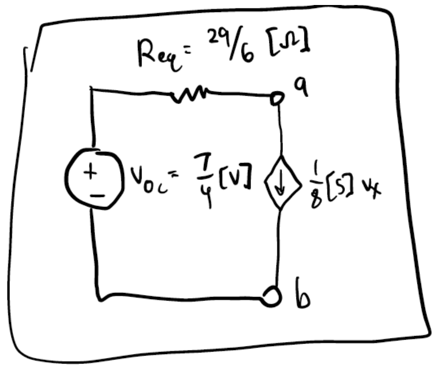
$$\text{KVL (1)}\quad -\frac{v_1}{10[\Omega]}\cdot 2[\Omega] - \frac{v_1}{10[\Omega]}\cdot 4[\Omega] + v_x = 0$$

$v_1 = \frac{20}{29}[V]$, $v_x = \frac{12}{29}[V]$, $i_T = \frac{6}{29}[A]$

$$R_{eq} = \frac{v_T}{i_T} = \frac{1[V]}{6/29[A]} = \boxed{\frac{29}{6}\ [\Omega]}$$

Room for extra work

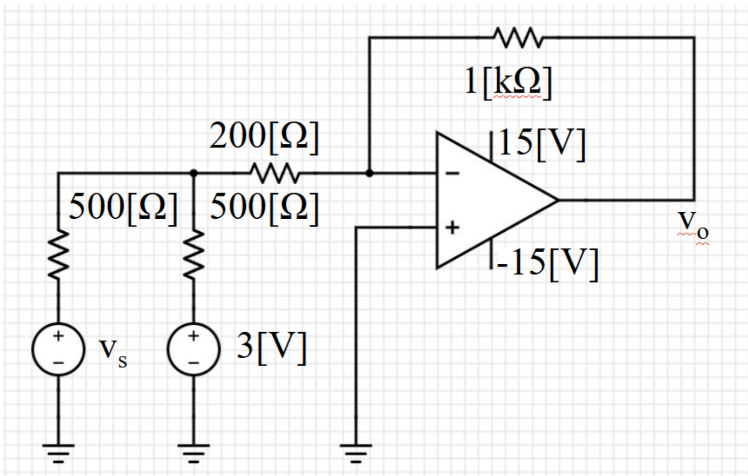
a)



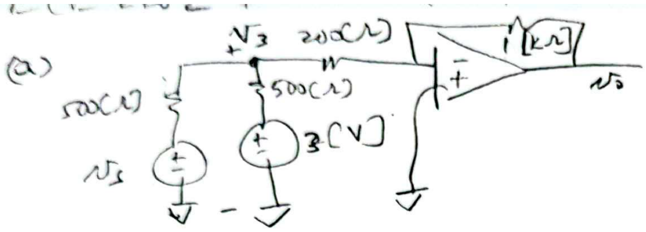
b) Resistor for max power $= R_{TH} = R_{eq} = \boxed{29/6\ [\Omega]}$

Draw Thevenin equivalent for Voc/Isc? (2 pts)
Draw Req/Isc equivalent? (2 pts)
Setup equations Voc/Isc – right/wrong (9 pts)
Setup equations Req/Isc – right/wrong (6 pts)
Values correct (Voc/Isc/Req) (1 pt)
Part b (concept – value should be same as Req) (2 pt)
Units (2 pts) and organization (1 pt)

3.{25 Points} Refer to the figure below.
a) Determine the output voltage (v_o) in the circuit below in terms of v_s.
b) Specify the linear range for v_s.
Label/organization 2 pts for node labels, 2 units
Part a) 10 pts for setup equations, 3 pts for simplified solutions
Part b) 4 pts for setup, 2pts for correct answer
2 pts freebie



(a)



① $\dfrac{v_s - v_3}{500(\Omega)} + \dfrac{3[V] - v_3}{500(\Omega)} = \dfrac{v_3}{200[\Omega]} \rightarrow v_3\left(\dfrac{1}{200} + \dfrac{1}{500} + \dfrac{1}{500}\right) = \dfrac{v_s}{500} + \dfrac{3}{500}$

② $\dfrac{v_3}{200(\Omega)} = \dfrac{0 - v_o}{1[k\Omega]} \rightarrow v_3 = \dfrac{-200\, v_o}{1{,}000}$

simplify ① further: $\boxed{v_3 = 0.222\, v_s + 0.667}$

Substitute into ②: $v_o = -\dfrac{1000}{200}(0.222 v_s + 0.1667)$

$\boxed{v_o = -1.111\, v_s - 3.333}$

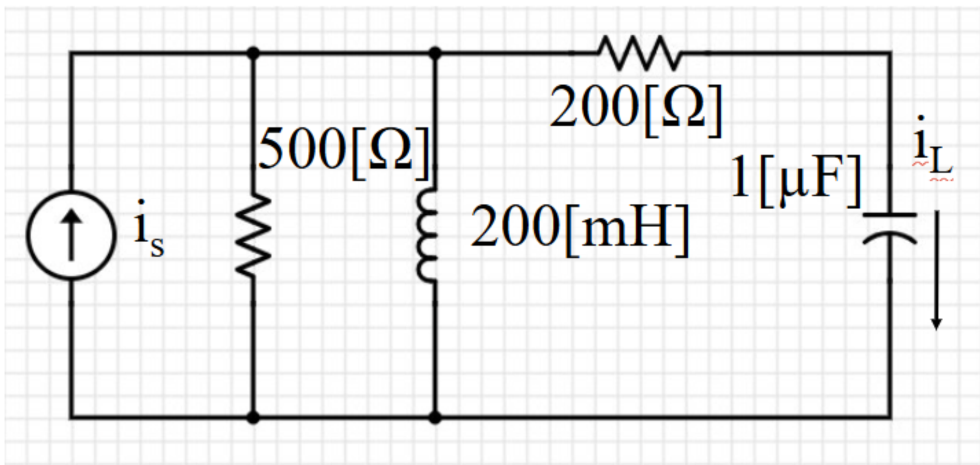
(b) $-15[V] \leq -1.111\, v_s - 3.333 \leq 15[V]$

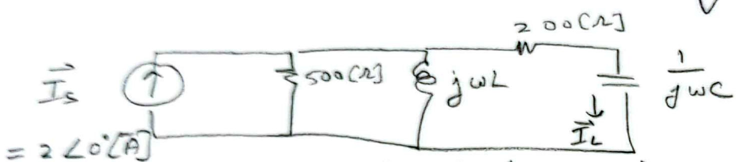
$-11.667 \leq -1.111\, v_s \leq 18.333[V]$

$\boxed{10.5011[V] \geq v_s \geq -16.5001[V]}$

4. {25 Points} Find $i_L(t)$ in the circuit below given that i_s=2*cos(2000[rad/s]t) [A].

3 pts for units/labels
7 pts for phasor domain conversion
Solution method (4pts) and final answer (3pts)
7 pts for conversion back to time domain
1 pts attempt





$$j\omega L = j(2000)(0.2[H]) = j400[\Omega]$$

$$\frac{1}{j\omega C} = \frac{1}{j(2000)(1[\mu F])} = -j500[\Omega]$$

$$500[\Omega] \,||\, j400[\Omega] = 195.12 + j243.9\ [\Omega] = \vec{Z}_1$$

Current divider: $\vec{I}_L = \vec{I}_s \left(\frac{\vec{Z}_1}{\vec{Z}_1 + (200 - j500)}\right)$

$$\vec{I}_L = 0.132 + j1.3201\ [A].$$

$$= 1.3267 \cos\left(2000\left[\frac{rad}{s}\right]t + 84.2894°\right)$$

Room for extra work